Nonlinear Dynamics & Chaos HW3

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Technion

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1 Coexistence in the two-species competition model

1.1 fixed points

at the fixed point

$$x(1 - x - ay) = 0 \qquad\qquad y(1 - y - ax) = 0 \tag{1}$$

which gives the fixed points, by guessing smartly

$$(0, 0) \quad (1, 0) \quad (0, 1) \tag{2}$$

but all three of those represent extinction of one or both species. We're interested in coexistence, i.e when $x, y \neq 0$ so

$$x + ay = 1 \qquad\qquad ax + y = 1 \tag{3}$$

subtracting the two equations gives $x = y$ so the fixed point is

$$(x, y) = \left(\frac{1}{1+a}, \frac{1}{1+a}\right) \tag{4}$$

1.2 condition on a

The jacobina is

$$J = \begin{pmatrix} 1 - 2x - ay & -ax \\ -ay & 1 - 2y - ax \end{pmatrix} \tag{5}$$

and at each fixed point

$$J(0,0) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \tag{6}$$

$$J(0,1) = \begin{pmatrix} 1-a & 0 \\ -a & -1 \end{pmatrix} \tag{7}$$

$$J(1,0) = \begin{pmatrix} -1 & -a \\ 0 & 1-a \end{pmatrix} \tag{8}$$

The eigenvalues for $0, 0$ are both 1 so it is an unstable node for all $a > 0$. for the other points, the eigenvalues are -1 and $1 - a$, so they are stable nodes for $0 < a < 1$ and a saddle for $a > 1$. For the coexistence fixed point the jacobian is

$$J = \begin{pmatrix} 1 - \frac{2}{1+a} - \frac{a}{1+a} & -\frac{a}{1+a} \\ \frac{-a}{1+a} & 1 - \frac{2}{1+a} - \frac{a}{1+a} \end{pmatrix} = \frac{-1}{1+a} \begin{pmatrix} 1 & a \\ a & 1 \end{pmatrix} \tag{9}$$

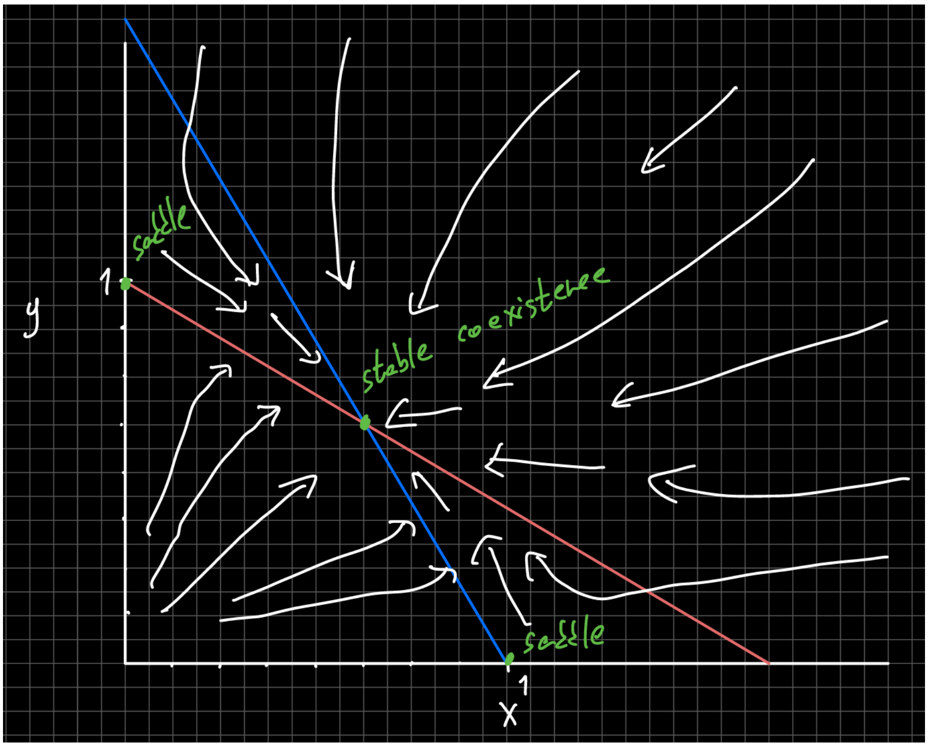
we can easily see the eigenvectors are $(1,1)$ and $(1,-1)$ so the eigenvalues are

$$\lambda_1 = \frac{-1}{1+a}(1+a) = -1 \tag{10}$$

$$\lambda_2 = \frac{-1}{1+a}(1-a) = \frac{a-1}{a+1} \tag{11}$$

so the coexistence is a stable node for $0 < a < 1$ and a saddle for $a > 1$.

1.3 phase space portrait



The x-nullclines are

$$x = 0, \quad 1 - x - ay = 0 \implies y = \frac{1-x}{a} \tag{12}$$

and the y-nullclines

$$y = 0, \quad 1 - y - ax = 0 \implies y = 1 - ax \tag{13}$$

their intersection point is

$$\left(\frac{1}{1+a}, \frac{1}{1+a}\right) \tag{14}$$

looking at the other fixed points, $(0,0)$ is a source, and both $(0,1)$ and $(1,0)$ are saddles, meaning that if we start at $x > 0, y > 0$ we eventually must end up at $\left(\frac{1}{1+a}, \frac{1}{1+a}\right)$.

2 A simple model of an epidemic

2.1 fixed points

for the fixed points

$$xky = 0 \qquad\qquad y\,(kx - l) = 0 \tag{15}$$

so all $(x, 0)$ is a line of fixed points, representing the case where there is no epidemic anymore.

2.2 trajectories

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\dot{y}}{\dot{x}} = \frac{kxy - ly}{-kxy} = -1 + \frac{l}{kx} \tag{16}$$

integrating

$$y = -x + \frac{l}{k}\ln x + C \tag{17}$$

so the conserved quantity is

$$H = x + y - \frac{l}{k}\ln x \tag{18}$$

so we can say

$$x + y - \frac{l}{k}\ln x = x_0 + y_0 - \frac{l}{k}\ln x_0 \tag{19}$$

therefore

$$y = y_0 + x_0 - x + \frac{l}{k}\ln\left(\frac{x}{x_0}\right) \tag{20}$$

2.3 sketch

2.3.1 nullclines

the nullclines for the xdot expression are

$$x = 0, \quad y = 0 \tag{21}$$

and for ydot

$$x = \frac{l}{k}, \quad y = 0 \tag{22}$$

and for every point $x, y > 0$ we have $\dot{x} = -kxy < 0$, so all the trajectories move left, and it's pretty simple to see that $\dot{y} > 0$ when $x > l/k$ and vice versa. So the trajectories point left and up to the right of $x = l/k$ and down to the left of it.

2.3.2 maximized y

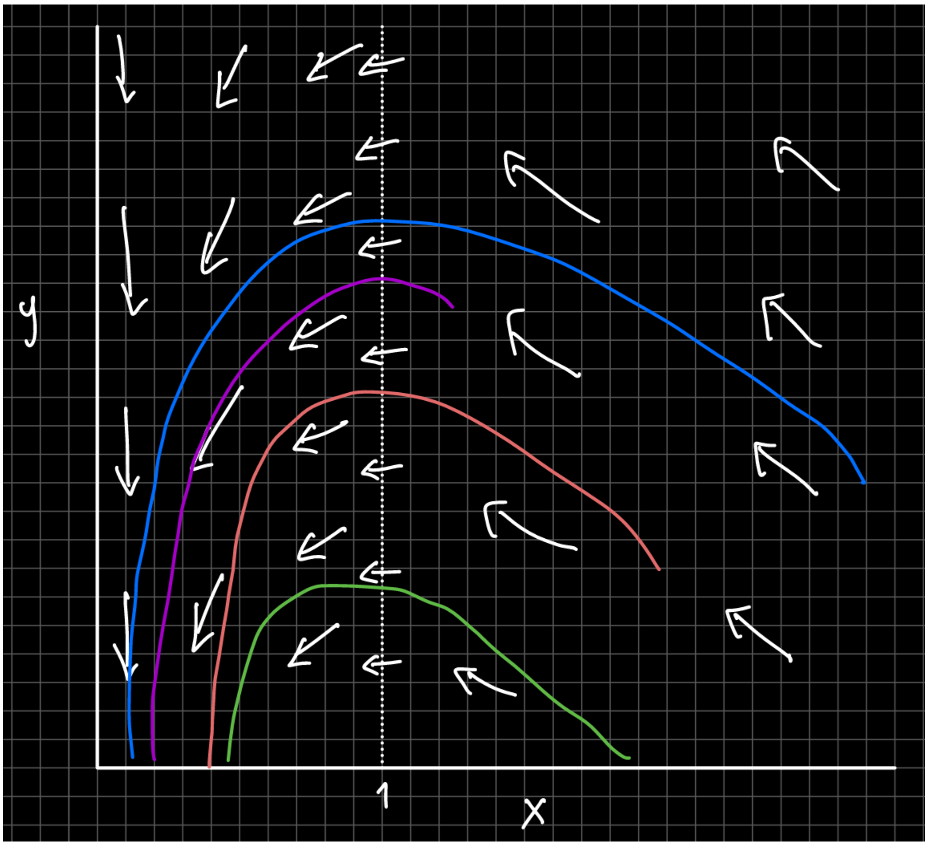
a maxima happens when $\dot{y} = 0$ so in our case

$$x = l/k \tag{23}$$

inserting into our expression for y, when $x_0 > l/k$

$$y_{max} = y_0 + x_0 - \frac{l}{k} + \frac{l}{k}\ln\left(\frac{l/k}{x_0}\right) \tag{24}$$

and when $x_0 < l/k$ y immediately decreases making y_0 the maximum.



2.4 Epidemic threshold

at $t = 0$

$$\dot{y}(0) = y_0 \, (kx_0 - l) \tag{25}$$

and since $y_0 > 0$

$$\dot{y}(0) > 0 \implies kx_0 - l > 0 \implies x_0 > \frac{l}{k} \tag{26}$$

using $R_0 = kx_0/l$ then the condition is

$$R_0 > 1 \tag{27}$$

lowering the susceptible pool can prevent an outbreak by lowering the ratio l/k, thus lowering R_0 below 1 further.

3 Phase portrait of the conservative pendulum

I'll define a helper variable

$$v = \dot{\theta} \tag{28}$$

so the equation becomes

$$\dot{v} = -\sin\theta \tag{29}$$

3.1 fixed points

at the fixed points

$$v = 0, \quad \sin\theta = 0 \tag{30}$$

so

$$(\theta^*, v^*) = (\pi n, 0) \tag{31}$$

next the jacobian is

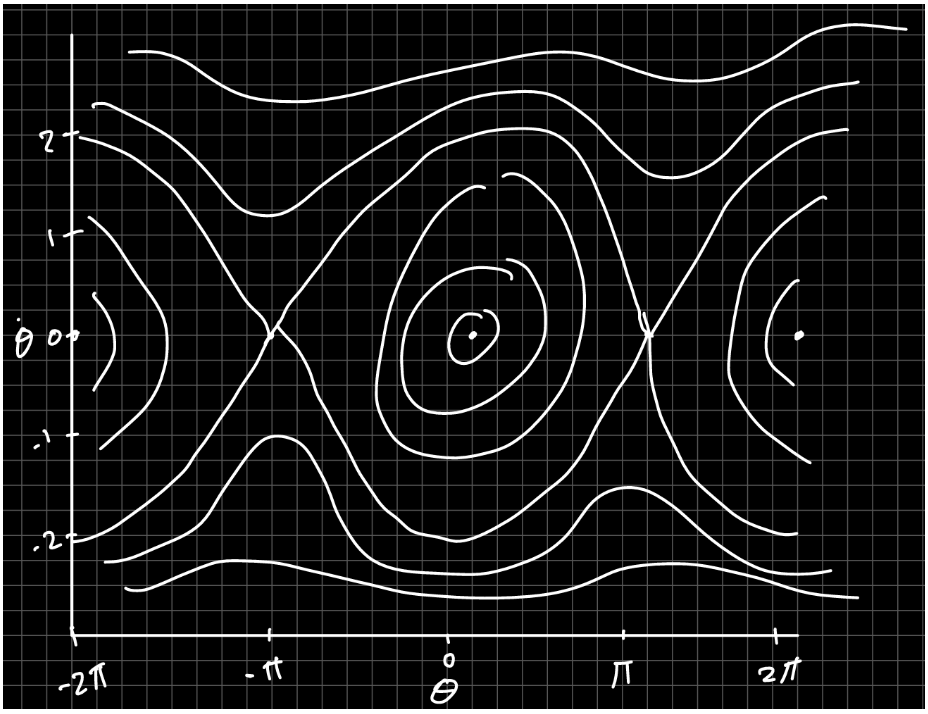
$$J = \begin{pmatrix} 0 & 1 \\ -\cos\theta & 0 \end{pmatrix} \tag{32}$$

we can split the fixed points into two groups, for even and odd n. Even n gives $\cos(\cdots) = 1$ therefore the eigenvalues are $\pm i$, so these are linear centers. Odd n gives $\cos(\cdots) = -1$ so the eigenvalues are ± 1, and since both eigenvalues are real but one is positive and the other is negative these are saddles.

3.2 genuine centers

Conservation of energy guarantees this. This can be shown explicitly by calculating the energy, but even a verbal argument I believe shows this. If we assume a trajectory which either spirals into or away from one such fixed point, it means the pendulum must be simultaneously moving away/toward from the fixed point physically and also gathering/losing velocity as it does so. This means energy is introduced/taken away from the system. Since this is impossible as we assume an isolated system, trajectories must be closed curves surrounding the fixed points, meaning they are genuine centers.

3.3 phase plane



The energy of the system is

$$E = \frac{1}{2}v^2 + 1 - \cos\theta \tag{33}$$

and the energy at the saddle points is

$$E((2m+1)\pi, 0) = 1 - \cos\left((2m+1)\,\pi\right) = 2 \tag{34}$$

so the separatrix is

$$\frac{1}{2}v^2 + 1 - \cos\theta = 2 \tag{35}$$

solving for v

$$v = \pm\sqrt{2\left(1 + \cos\theta\right)} \tag{36}$$

which can be written as

$$v = \pm 2\cos\left(\frac{\theta}{2}\right) \tag{37}$$

which represents a pendulum with just enough energy to reach the top of its motion with 0 velocity.

4 Period of a relaxation oscillation

4.1 Lienard variable

the lienard variable

$$y = \frac{\dot{x}}{k} + F(x) \tag{38}$$

so

$$\dot{x} = k\left(y - F(x)\right) \tag{39}$$

and

$$\dot{y} = \frac{\ddot{x}}{k} + F'(x)\dot{x} = \frac{\ddot{x} + k\left(x^2 - 4\right)\dot{x}}{k} \tag{40}$$

inserting the original equation

$$\dot{y} = \frac{1-x}{k} \tag{41}$$

so the system is

$$\dot{x} = k\left(y - \frac{x^3}{3} + 4x\right) \qquad \dot{y} = \frac{1-x}{k} \tag{42}$$

and we can notice that x is fast and y is slow.

4.2 slow parts

The stability if the fast part is determined by

$$\frac{\partial \dot{x}}{\partial x} = -kF'(x) = -k\left(x^2 - 4\right) \tag{43}$$

which is attracting when

$$F'(x) = x^2 - 4 > 0 \tag{44}$$

i.e

$$x < -2 \quad \text{or} \quad x > 2 \tag{45}$$

so we look at $x = \pm 2$ with

$$F(2) = -\frac{16}{3} \quad F(-2) = \frac{16}{3} \tag{46}$$

finding the other points at which we get these values

$$x^3 - 12x - 16 = 0 \implies (x-4)(x+2)^2 \tag{47}$$

so the trajectory jumps

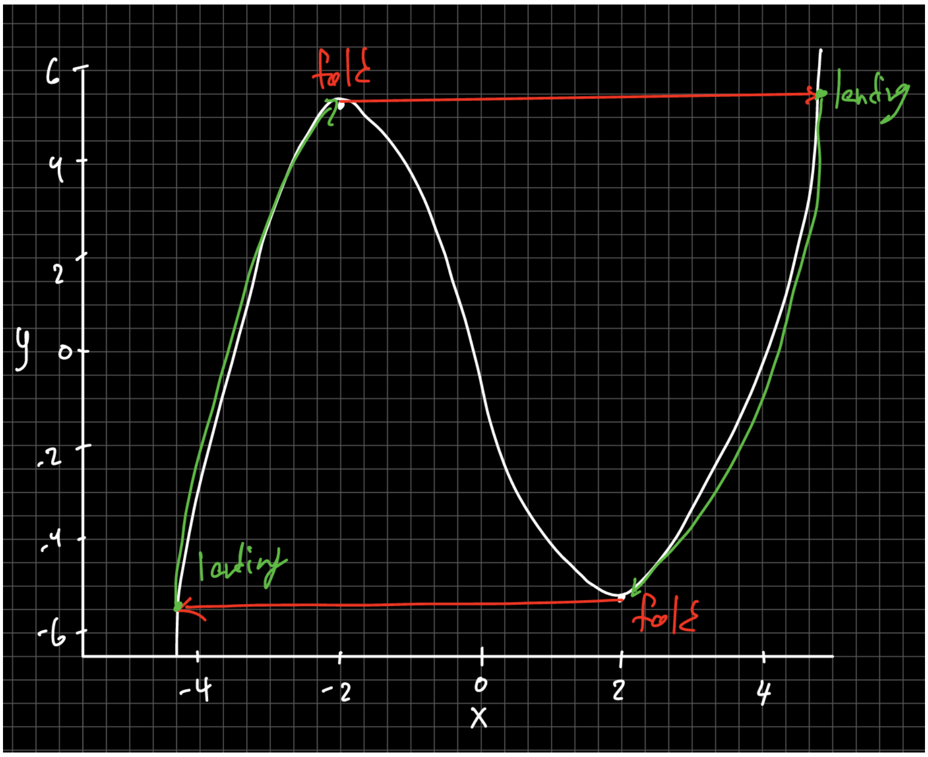
$$(-2, 16/3) \to (4, 16/3) \tag{48}$$

similarly

$$(2, -16/3) \to (-4, -16/3) \tag{49}$$

so the slow parts are

$$(-4, -16/3) \to (-2, 16/3) \qquad (4, 16/3) \to (2, -16/3) \tag{50}$$



4.3 period

on a slow branch

$$y \sim F(x) \tag{51}$$

so

$$\dot{y} \sim F'(x)\dot{x} = \left(x^2 - 4\right)\dot{x} \tag{52}$$

using $\dot{y} = \frac{1-x}{k}$

$$\dot{x} = \frac{1-x}{k\left(x^2-4\right)} \tag{53}$$

so

$$\mathrm{d}t = k\frac{x^2-4}{1-x}\,\mathrm{d}x \tag{54}$$

so

$$T = k\int_a^b \frac{x^2-4}{1-x} = k\left(-\frac{x^2}{2} - x + 3\ln|x-1|\right)\Big|_a^b \tag{55}$$

on the right branch x goes from 4 to 2 so

$$T_R = k\left(-\frac{x^2}{2} - x + 3\ln|x-1|\right)\Big|_4^2 = k\left(8 - 3\ln 3\right) \tag{56}$$

and on the left x goes from -4 to -2

$$T_L = k\left(-\frac{x^2}{2} - x + 3\ln|x-1|\right)\Big|_{-4}^{-2} = k\left(4 - 3\ln\frac{5}{3}\right) \tag{57}$$

so the period time is approximately

$$T = T_R + T_L = k\,(8 - 3\ln 3) + k\left(4 - 3\ln\frac{5}{3}\right) = k\,(12 - 3\ln 5) \tag{58}$$

5 Secular terms in weakly non-linear oscillators

the equation is

$$\ddot{x} + x = -\varepsilon \left(a_2 x^2 + a_3 x^3 \right), \quad 0 < \varepsilon \ll 1 \tag{59}$$

5.1 two time expansion

we define

$$\tau = t \quad T = \varepsilon t \tag{60}$$

so that

$$\partial_t = \partial_\tau + \varepsilon \partial_T \tag{61}$$

and

$$\partial_t^2 = \partial_\tau^2 + 2\varepsilon \partial_\tau \partial_T + O\left(\varepsilon^2\right) \tag{62}$$

we expand

$$x = x_0(\tau, T) + \varepsilon x_1(\tau, T) + O(\varepsilon^2) \tag{63}$$

substituting into the equation

$$\left(\partial_\tau^2 + 2\varepsilon \partial_\tau \partial_T\right)\left(x_0 + \varepsilon x_1\right) + x_0 + \varepsilon x_1 = -\varepsilon \left(a_2 x_0^2 + a_3 x_0^3\right) + O(\varepsilon^2) \tag{64}$$

and at $O(1)$

$$\partial_\tau^2 x_0 + x_0 = 0 \tag{65}$$

whose solution can be written as

$$x_0\left(\tau, T\right) = r(T)\cos\psi, \quad \psi = \tau + \phi(T) \tag{66}$$

where r is the amplitude and ϕ is the phase. At order $O(\varepsilon)$

$$\partial_\tau^2 x_1 + x_1 = -2\partial_\tau \partial_T x_0 - a_2 x_0^2 - a_3 x_0^3 \tag{67}$$

5.2 higher harmonics

Starting from

$$x_0 = r\cos\psi \tag{68}$$

then

$$\partial_T x_0 = \partial_T r \cos\psi - r\partial_T \phi \sin\psi \tag{69}$$

then

$$\partial_\tau \partial_T x_0 = -\partial_T r \sin\psi - r\partial_T \phi \cos\psi \tag{70}$$

so

$$-2\partial_\tau \partial_T x_0 = \partial_T r \sin\psi + r\partial_T \phi \cos\psi \tag{71}$$

and using the identities

$$\cos^2\psi = \frac{1}{2}\left(1+\cos 2\psi\right) \qquad \cos^3\psi = \frac{1}{4}\left(3\cos\psi + \cos 3\psi\right) \tag{72}$$

we get

$$-a_2 x_0^2 = -\frac{a_2 r^2}{2} - \frac{a_2 r^2}{2}\cos 2\psi \tag{73}$$

and

$$-a_3 x_0^3 = -\frac{3a_3 r^3}{4}\cos\psi - \frac{a_3 r^3}{4}\cos 3\psi \tag{74}$$

combining everything

$$\partial_\tau^2 x_1 + x_1 = \partial_T r \sin\psi + \left(2r\partial_T\phi - \frac{3a_3 r^3}{4}\right)\cos\psi - \frac{a_2 r^2}{2}\left(1+\cos 2\psi\right) - \frac{a_3 r^3}{4}\cos 3\psi \tag{75}$$

5.3 resonant terms

The resonant terms are $2\partial_T r\sin\psi$ and $\left(2r\partial_T\phi - \frac{3a_3 e^3}{4}\right)\cos\psi$. To eliminate secular growth their coefficients must vanish so

$$\partial_T r = 0 \implies r(T) = r_0 \tag{76}$$

and

$$\partial_T\phi = \frac{3a_3 r^2}{8} \implies \phi(T) = \phi_0 + \frac{3}{8}a_3 r_0^2 T \tag{77}$$

inserting $T = \varepsilon t$

$$x(t) \simeq r_0 \cos\left(\left(1 + \frac{3}{8}\varepsilon a_3 r_0^2\right)t + \phi_0\right) \tag{78}$$

5.4 duffing oscillator

in the duffing oscillator

$$a_2 = 0, \quad a_3 = 1 \tag{79}$$

giving

$$\partial_T r = 0, \quad \partial_T\phi = \frac{3}{8}r^2 \tag{80}$$

A nonzero quadratic nonlinearity a_2 doesn't change the leading order slow equations because

$$x_0^2 = \frac{r^2}{2}\left(1+\cos 2\psi\right) \tag{81}$$

only has a constant term and a second harmonic, and since it doesn't have the natural frequency of the oscillator, it's not resonant.

5.5 averaged equations

The averaged equations from the notes are

$$\dot{A} = \varepsilon\left\langle h\sin\theta\right\rangle, \quad A\dot{\phi} = \varepsilon\left\langle h\cos\theta\right\rangle, \quad \langle\cdot\rangle = \frac{1}{2\pi}\int_{-\pi}^{\pi}(\cdot)\,\mathrm{d}\theta \tag{82}$$

inserting

$$x = A\cos\theta, \quad \dot{x} = -A\sin\theta \tag{83}$$

where $\theta = t + \phi(t)$ into h

$$h = A^3 \cos\theta \sin^2\theta \tag{84}$$

making the amplitude equation

$$\dot{A} = \varepsilon A^3 \left\langle \cos\theta \sin^3\theta \right\rangle \tag{85}$$

but the average over a period of the trig function zeroes

$$\dot{A} = 0 \tag{86}$$

and for the phase

$$\begin{aligned} A\dot{\phi} &= \varepsilon A^3 \left\langle \cos^2\theta \sin^2\theta \right\rangle && (87)\\ &= \varepsilon A^3 \left\langle \frac{1}{4}\sin^2 2\theta \right\rangle && (88)\\ &= \frac{\varepsilon A^3}{4} \cdot \frac{1}{2} && (89)\\ &= \frac{\varepsilon A^3}{8} && (90) \end{aligned}$$

therefore

$$\dot{\phi} = \frac{\varepsilon A^2}{8} \tag{91}$$

5.6 solution

For the amplitude,

$$A = A_0 \tag{92}$$

to be determined by the initial condition. For the phase,

$$\phi = \phi_0 + \frac{\varepsilon A^2}{8} t \tag{93}$$

so the averaged solution is

$$x(t) \simeq A_0 \cos\left(\left(1 + \frac{\varepsilon A^2}{8}\right) t + \phi_0\right) \tag{94}$$

in leading order, and the inital conditions have

$$x(0) = A_0 \cos\phi_0, \quad \dot{x}(0) = -A_0 \sin\phi_0 \tag{95}$$

which we can combine and rearrange to

$$A_0 \simeq \sqrt{x(0)^2 + \dot{x}(0)^2} \tag{96}$$

So the slow flow predicts that the amplitude remains constant at its initial value. There is no attracting value of A meaning the system also doesn't have a limit cycle. Looking at the frequency

$$\omega = 1 + \frac{\varepsilon A_0^2}{8} \tag{97}$$

we see it only depends on the initial amplitude, so once that is set, the frequency is also constant.